

1Title : Question Paper

FF No. 868

Reg.No.						
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**Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE – 411037.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)**

Examination: ESE

Year: F.Y.B.Tech.

Branch: All

Subject: Calculus and Statistics

Subject Code: HS1076

Max. Marks: 60

Total Pages of Question Paper:-

Day & Date: Tuesday, 18.2.25

Time:

Instructions to Candidate

1. All questions are compulsory.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right indicate full marks.

Q. No.		Max marks
Q. 1.A	Solve(Any one)	03 Marks
	Ratio test/comparison test/interval of convergence	
Q. 1.B	Taylor/Maclaurin's series	02 Marks
Q. 2.	Solve (Any one)	05 marks
	Chain Rule/Jacobian/maxima minima/Errors & approximations	
Q. 3.	Solve (Any one)	05 marks
	Directional Derivatives/Gradient/divergence/curl	
Q. 4.	Solve (Any three)	15 marks
	Evaluation of double/triple integrals using cartesian/polar/spherical/cylindrical coordinates	
Q 5.A	Solve (Any two)	12 marks
	Mean-mode-median of grouped frequency distribution/Coefficient of skewness/histogram/boxplot	
Q 5.B	Moments/coefficient of skewness-kurtosis based on moments	03 marks
Q. 6.A	Solve (Any two)	12 marks
	Problems on pmf/pdf/cdf/expectation/variance of discrete and continuous random variables	
Q6.B	Bivariate(joint) probability distribution of discrete random variable.	03 marks